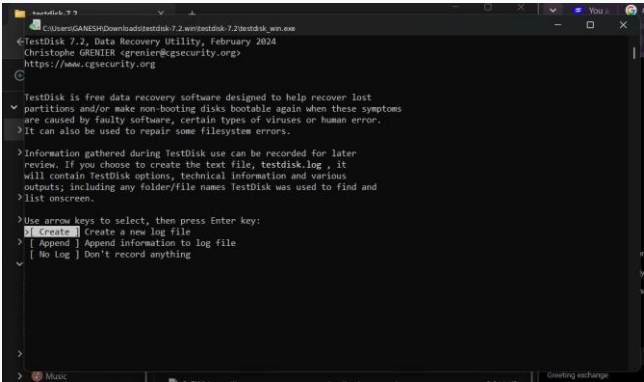


Ex.No.02 - TestDisk Forensic Recovery

Experiment Name	TestDisk Forensic Recovery
Aim	To analyse a storage device using TestDisk, identify the required partition, verify its file structure, and perform partition recovery operations.
Tool Used	TestDisk 7.2, Windows, 2 GB Virtual Test Disk, NTFS File System
Safety Precaution	The experiment was performed on a 2 GB virtual test disk instead of the actual laptop storage to avoid accidental modification or data loss on the actual system disk.

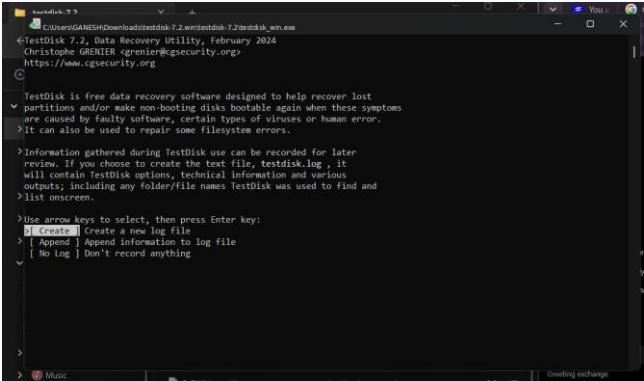
Experiment Focus: Disk analysis, partition identification, Quick Search, partition verification, file listing, Deeper Search, partition information writing, and final verification.

Output 1 - Create Log Screen



TestDisk was launched with administrative privileges and the Create option was selected to create a new log file for the recovery session.

Output 2 - Storage Device Selection



The available storage devices were displayed. The 2 GB virtual test disk was selected and Proceed was chosen to continue.

Forensic significance: The log and storage-device screens establish the controlled virtual-disk recovery environment.

Output 3 - Partition Table Type Selection

```
C:\Users\GANESH\Downloads>testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - Msft Virtual Disk

Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac   ] Apple partition map (legacy)
[None  ] Non partitioned media
[Sun   ] Sun Solaris partition
[XBox  ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

The Intel/PC (MBR) partition table type was selected for the virtual test disk so that TestDisk could interpret its partition structure.

Output 4 - Analyse Disk

```
C:\Users\GANESH\Downloads>testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - Msft Virtual Disk

Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac   ] Apple partition map (legacy)
[None  ] Non partitioned media
[Sun   ] Sun Solaris partition
[XBox  ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

The Analyse option was selected to examine the current partition structure and search for partitions on the virtual storage device.

Forensic significance: Partition-table selection and disk analysis determine how TestDisk interprets and examines the storage structure.

Output 5 - Quick Search Results

```
TestDisk 7.2
C:\Users\GAMESH\Downloads\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - GHS 261 255 63
Partition      Start      End      Size in sectors
L HPFS - NTFS   0 2 3    268 100 48  4188100 [TestDiskLab]

Structure: OK. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*-Primary bootable P*-Primary L-Logical E-Extended D-Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2144 MB / 2045 MiB
```

Quick Search scanned the virtual disk and identified the NTFS partition. The detected partition structure was reviewed before verification.

Output 6 - Partition Verification

```
TestDisk 7.2
C:\Users\GAMESH\Downloads\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

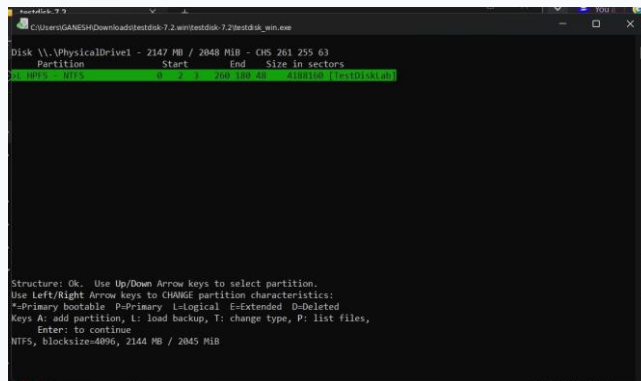
Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - GHS 261 255 63
Partition      Start      End      Size in sectors
L HPFS - NTFS   0 2 3    268 100 48  4188100 [TestDiskLab]

Structure: OK. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*-Primary bootable P*-Primary L-Logical E-Extended D-Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2144 MB / 2045 MiB
```

The detected partition was displayed as L HPFS - NTFS [TestDiskLab]. The partition entry was selected for verification.

Forensic significance: Quick Search and partition verification identify and validate the NTFS partition before recovery actions.

Output 7 - File Listing



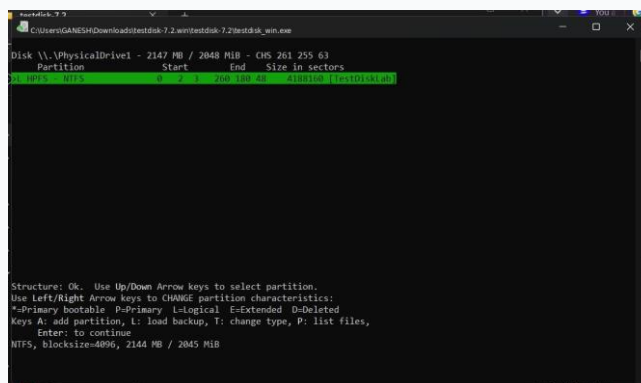
```
C:\Users\GANESH\Downloads\testdisk-7.2\testdisk_win.exe

Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - CHS 261 255 63
Partition      Start      End      Size in sectors
0 1975  - NTFS    0 2 1  200 100 48  4100160 [1st1013kub]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*-Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2144 MB / 2045 MiB
```

The P key was used to list the files present in the detected partition. TestDisk displayed the directory structure and accessible files.

Output 8 - Deeper Search Results



```
C:\Users\GANESH\Downloads\testdisk-7.2\testdisk_win.exe

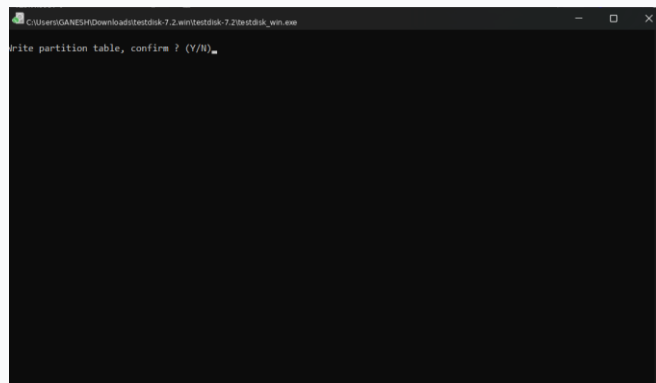
Disk \\.\PhysicalDrive1 - 2147 MB / 2048 MiB - CHS 261 255 63
Partition      Start      End      Size in sectors
0 1975  - NTFS    0 2 1  200 100 48  4100160 [1st1013kub]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*-Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2144 MB / 2045 MiB
```

The Deeper Search option was performed to conduct a more thorough search of the storage device. The detected partition was verified again.

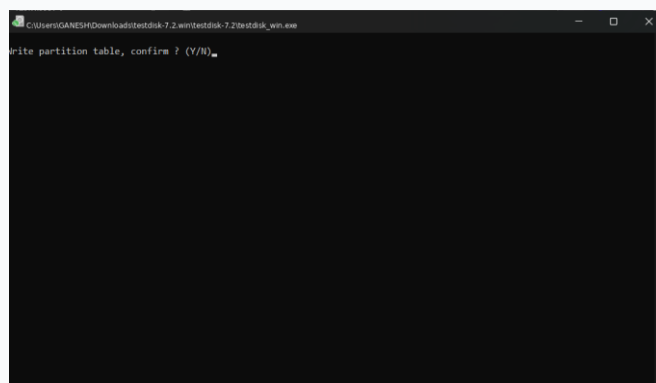
Forensic significance: File listing and Deeper Search provide additional evidence that the partition structure and contents are accessible.

Output 9 - Write Partition Information



After verifying the correct partition, the Write option was selected. TestDisk displayed a confirmation prompt before writing the partition information.

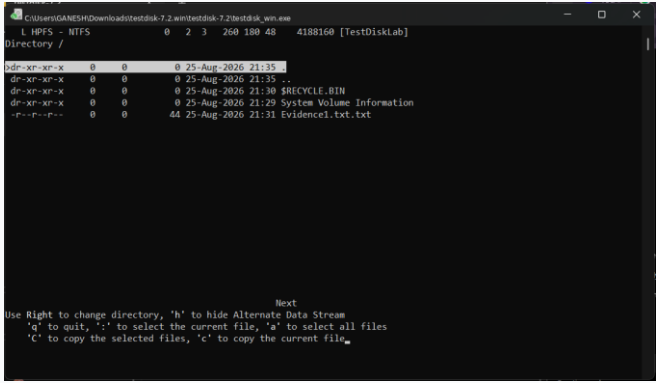
Output 10 - Write Confirmation / Exit



The partition-table write operation was confirmed using Y. TestDisk then displayed the restart requirement and the application was exited using Quit.

Forensic significance: The Write and confirmation screens document the partition-recovery action and its completion.

Output 11 - Final File Listing



The final file-listing screen was used to verify that the detected NTFS partition and its file structure were accessible.

Observation

The virtual storage device was successfully analysed using TestDisk. The NTFS partition [TestDiskLab] was detected, verified, searched more deeply, and its file structure was accessed.

Result

TestDisk successfully analysed the virtual storage device, identified the NTFS partition, verified its file structure, and completed the partition recovery workflow.

Conclusion

The experiment demonstrated the practical use of TestDisk for forensic partition analysis and recovery. Quick Search, Deeper Search, verification, file listing, and partition-table writing were performed in sequence.

Key forensic keywords
TestDisk MBR NTFS Quick Search Deeper Search Partition Recovery File Listing Partition Table Virtual Disk